



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
Canada WHMIS 2015 which includes the amended Hazardous Products Act (HPA) and the Hazardous Products Regulation (HPR)

Revision date 28-Aug-2024

Revision Number 1

## 1. Identification

### Product identifier

Product Name Sequential Elution Reagent 3

### Other means of identification

Catalog Number(s) 10010275

### Recommended use of the chemical and restrictions on use

Recommended use Laboratory chemicals

Restrictions on use No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
1329 Meyerside Drive  
Mississauga, ON L5T 1C9  
Canada

#### Technical Service

1-800-361-1808  
support@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

Acute toxicity - Oral

Category 4

### Label elements

#### **Warning**

#### **Hazard statements**

Harmful if swallowed



**Precautionary Statements - Prevention**

Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product

**Precautionary Statements - Response****Ingestion**

IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell  
Rinse mouth

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

No information available.

**Unknown acute toxicity**

0 % of the mixture consists of ingredient(s) of unknown acute oral toxicity

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

| Chemical name   | CAS No.  | Weight-% | Hazardous Material<br>Information Review Act<br>registry number<br>(HMIRA registry #) | Date HMIRA filed and<br>date exemption granted<br>(if applicable) |
|-----------------|----------|----------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Ethylene glycol | 107-21-1 | 50 - 100 | -                                                                                     |                                                                   |

**4. First-aid measures****Description of first aid measures**

|                       |                                                                                                                                                                  |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b> | Show this safety data sheet to the doctor in attendance.                                                                                                         |
| <b>Inhalation</b>     | Remove to fresh air.                                                                                                                                             |
| <b>Eye contact</b>    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a physician.                                          |
| <b>Skin contact</b>   | Wash skin with soap and water.                                                                                                                                   |
| <b>Ingestion</b>      | Do NOT induce vomiting. Clean mouth with water and drink afterwards plenty of water.<br>Never give anything by mouth to an unconscious person. Call a physician. |

**Most important symptoms and effects, both acute and delayed**

**Symptoms** No information available.

**Indication of any immediate medical attention and special treatment needed**

**Note to physicians** Treat symptomatically.

## 5. Fire-fighting measures

**Suitable Extinguishing Media** Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Unsuitable extinguishing media** No information available.

**Specific hazards arising from the chemical** None known.

**Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Special protective equipment for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## 6. Accidental release measures

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** See section 8 for more information.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

## 7. Handling and storage

**Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Keep out of the reach of children. Store according to product and label instructions.

## 8. Exposure controls/personal protection

**Control parameters**

**Exposure Limits**

| Chemical name               | Alberta                        | British Columbia                                                                                             | Ontario                                                   | Quebec                                            |
|-----------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------|
| Ethylene glycol<br>107-21-1 | Ceiling: 100 mg/m <sup>3</sup> | TWA: 10 mg/m <sup>3</sup><br>STEL: 20 mg/m <sup>3</sup><br>Ceiling: 100 mg/m <sup>3</sup><br>Ceiling: 50 ppm | TWA: 25 ppm<br>STEL: 50 ppm<br>STEL: 10 mg/m <sup>3</sup> | Ceiling: 50 ppm<br>Ceiling: 127 mg/m <sup>3</sup> |

**Appropriate engineering controls**

**Engineering controls**                      Showers  
                                                          Eyewash stations  
                                                          Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection**                      Wear safety glasses with side shields (or goggles).

**Hand protection**                              Wear suitable gloves.

**Skin and body protection**                      Wear suitable protective clothing.

**Respiratory protection**                      No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations**                      Handle in accordance with good industrial hygiene and safety practice.

**9. Physical and chemical properties****Information on basic physical and chemical properties**

**Physical state**                                      Liquid  
**Appearance**                                      aqueous solution  
**Color**                                                  colorless  
**Odor**                                                  Odorless  
**Odor threshold**                                      No information available

| <b><u>Property</u></b>                         | <b><u>Values</u></b> | <b><u>Remarks • Method</u></b> |
|------------------------------------------------|----------------------|--------------------------------|
| <b>pH</b>                                      |                      | None known                     |
| <b>Melting point / freezing point</b>          | No data available    | None known                     |
| <b>Initial boiling point and boiling range</b> | No data available    | None known                     |
| <b>Flash point</b>                             | 111 °C / 231.8 °F    |                                |
| <b>Evaporation rate</b>                        | No data available    | None known                     |
| <b>Flammability</b>                            | No data available    | None known                     |
| <b>Flammability Limit in Air</b>               |                      | None known                     |
| <b>Upper flammability or explosive limits</b>  | No data available    |                                |
| <b>Lower flammability or explosive limits</b>  | No data available    |                                |
| <b>Vapor pressure</b>                          | No data available    | None known                     |
| <b>Relative vapor density</b>                  | No data available    | None known                     |
| <b>Relative density</b>                        | 1.06                 | None known                     |
| <b>Water solubility</b>                        | Miscible in water    |                                |
| <b>Solubility in other solvents</b>            | No data available    | None known                     |
| <b>Partition coefficient</b>                   | No data available    | None known                     |
| <b>Autoignition temperature</b>                | 398 °C / 748.4 °F    |                                |
| <b>Decomposition temperature</b>               |                      | None known                     |
| <b>Kinematic viscosity</b>                     | No data available    | None known                     |
| <b>Dynamic viscosity</b>                       | No data available    | None known                     |

Other information

|                      |                 |
|----------------------|-----------------|
| Explosive properties | Not applicable. |
| Oxidizing properties | Not applicable. |
| Softening point      | Not applicable  |
| Molecular weight     | Not applicable  |
| VOC content          | Not applicable  |

**10. Stability and reactivity**

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

**11. Toxicological information**Information on likely routes of exposure**Product Information**

|              |                                                                                                              |
|--------------|--------------------------------------------------------------------------------------------------------------|
| Inhalation   | May be harmful if inhaled.                                                                                   |
| Eye contact  | Specific test data for the substance or mixture is not available.                                            |
| Skin contact | Specific test data for the substance or mixture is not available.                                            |
| Ingestion    | Specific test data for the substance or mixture is not available Harmful if swallowed (based on components). |

Symptoms related to the physical, chemical and toxicological characteristics

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

Acute toxicity**Numerical measures of toxicity**

No information available

**The following values are calculated based on chapter 3.1 of the GHS document**

|                               |                 |
|-------------------------------|-----------------|
| ATEmix (oral)                 | 833.30 mg/kg    |
| ATEmix (dermal)               | 17,666.70 mg/kg |
| ATEmix (inhalation-dust/mist) | 6.25 mg/l       |

**Unknown acute toxicity**

0 % of the mixture consists of ingredient(s) of unknown acute oral toxicity

**Component Information**

| Chemical name               | Oral LD50            | Dermal LD50           | Inhalation LC50        |
|-----------------------------|----------------------|-----------------------|------------------------|
| Ethylene glycol<br>107-21-1 | = 4700 mg/kg ( Rat ) | = 10600 mg/kg ( Rat ) | > 2.5 mg/L ( Rat ) 6 h |

Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                          |                                                                   |
|------------------------------------------|-------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Based on available data, the classification criteria are not met. |
| <b>Serious eye damage/eye irritation</b> | Based on available data, the classification criteria are not met. |
| <b>Respiratory or skin sensitization</b> | Based on available data, the classification criteria are not met. |
| <b>Germ cell mutagenicity</b>            | Based on available data, the classification criteria are not met. |
| <b>Carcinogenicity</b>                   | Based on available data, the classification criteria are not met. |
| <b>Reproductive toxicity</b>             | Based on available data, the classification criteria are not met. |
| <b>STOT - single exposure</b>            | Based on available data, the classification criteria are not met. |
| <b>STOT - repeated exposure</b>          | Based on available data, the classification criteria are not met. |
| <b>Target organ effects</b>              | Respiratory system, Eyes, Skin, Central nervous system.           |
| <b>Aspiration hazard</b>                 | Based on available data, the classification criteria are not met. |

## 12. Ecological information

### Ecotoxicity

| Chemical name               | Algae/aquatic plants                                                     | Fish                                                                                                                                                                                                                                                                                                                                           | Toxicity to microorganisms | Crustacea                                     |
|-----------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------------------------------------------|
| Ethylene glycol<br>107-21-1 | EC50: 6500 - 13000mg/L<br>(96h, <i>Pseudokirchneriella subcapitata</i> ) | LC50: =41000mg/L (96h, <i>Oncorhynchus mykiss</i> )<br>LC50: 14 - 18mL/L (96h, <i>Oncorhynchus mykiss</i> )<br>LC50: =27540mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =40761mg/L (96h, <i>Oncorhynchus mykiss</i> )<br>LC50: 40000 - 60000mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: =16000mg/L (96h, <i>Poecilia reticulata</i> ) | -                          | EC50: =46300mg/L (48h, <i>Daphnia magna</i> ) |

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

### Component Information

| Chemical name               | Partition coefficient |
|-----------------------------|-----------------------|
| Ethylene glycol<br>107-21-1 | -1.36                 |

**Other adverse effects** No information available.

## 13. Disposal considerations

### Disposal methods

|                                            |                                                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused products</b> | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| <b>Contaminated packaging</b>              | Do not reuse empty containers.                                                                                  |

## 14. Transport information

|                    |               |
|--------------------|---------------|
| <b><u>TDG</u></b>  | Not regulated |
| <b><u>DOT</u></b>  | Not regulated |
| <b><u>MEX</u></b>  | Not regulated |
| <b><u>IATA</u></b> | Not regulated |
| <b><u>IMDG</u></b> | Not regulated |

## 15. Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

#### International Inventories

Contact supplier for inventory compliance status

## 16. Other information

|                    |                         |                       |                           |                                           |
|--------------------|-------------------------|-----------------------|---------------------------|-------------------------------------------|
| <b><u>NFPA</u></b> | <b>Health hazards</b> 2 | <b>Flammability</b> 1 | <b>Instability</b> 0      | <b>Physical and chemical properties -</b> |
| <b><u>HMIS</u></b> | <b>Health hazards</b> 2 | <b>Flammability</b> 1 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X              |

### Key or legend to abbreviations and acronyms used in the safety data sheet

#### **Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |

#### **Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGl(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database

International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 28-Aug-2024

**Revision Note** Significant changes throughout SDS. Review all sections.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**